Time Series Study Sheet

Sarra Lajnef
Based on: Applied Time Series Analysis (Terence C. Mills), Course GLT-polito

April 2, 2026

Contents

1 Part 1: Time Series and Their Features **4**
1.1 Fundamental Definitions . . . 4
1.2 Key Features of Time Series . . . 4
1.2.1 Autocorrelation and Periodic Movements . . . 4
1.2.2 Seasonality . . . 4
1.2.3 Trends . . . 4
1.2.4 Stationarity and Nonstationarity . . . 4
1.2.5 Detrending via Iterated Differencing . . . 7
1.2.6 Deterministic Trend Decomposition . . . 7
1.2.7 ACF and PACF . . . 8
1.2.8 Ergodicity in the Mean . . . 10
1.2.9 Seasonality and Frequency Domain Analysis . . . 11
1.2.10 Custom Spectrum: DCT-II with Standardization . . . 12
1.2.11 Deterministic Seasonality Decomposition . . . 13
1.2.12 Distributional Analysis and Normality . . . 14
1.2.13 Last Example Of The lab 1 (see it) . . . 14
1.2.14 Volatility . . . 15
1.3 Natural Constraints . . . 15
1.4 Multivariate Features . . . 15

2 Part 2: Transforming Time Series **16**
2.1 Distributional Transformations . . . 16
2.1.1 Logarithmic Transformation . . . 16
2.1.2 Box-Cox Transformation . . . 16
2.1.3 Transformations for Proportions . . . 17
2.1.4 Handling Negative Values . . . 17
2.2 Stationarity Inducing Transformations . . . 17
2.2.1 First Differencing . . . 17
2.2.2 Seasonal Differencing . . . 17
2.2.3 The Shift Operator . . . 18
2.3 Decomposing a Time Series . . . 18
2.3.1 Additive vs. Multiplicative Models . . . 18
2.3.2 Smoothing Transformations (Moving Averages) . . . 18
2.3.3 Hodrick-Prescott (HP) Filter . . . 18

3 Part 3: ARMA Models for Stationary Time Series **19**
3.1 Intuitive Interpretation of AR and MA Components . . . 19
3.1.1 AR (Autoregressive): The Long-Term Memory . . . 19
3.1.2 MA (Moving Average): The Shock Absorber . . . 19
3.1.3 ARMA: The Combined Personality . . . 19
3.1.4 Summary Comparison . . . 19
3.2 Theoretical Foundations . . . 20
3.2.1 The Stochastic Process . . . 20
3.2.2 Stationarity (Covariance Stationarity) . . . 20
3.2.3 The White Noise Process . . . 20
3.3 Autoregressive Processes: $AR(p)$ . . . 20
3.3.1 The $AR(1)$ Model: Detailed Derivation . . . 20
3.3.2 The General $AR(p)$ and Yule-Walker Equations . . . 20
3.4 Moving Average Processes: $MA(q)$ . . . 21
3.4.1 The $MA(1)$ Model . . . 21
3.4.2 Invertibility . . . 21
3.4.3 General $MA(q)$ Autocorrelation . . . 21
3.5 The $ARMA(p, q)$ Process . . . 21
3.5.1 General Definition . . . 21
3.5.2 The Partial Autocorrelation Function (PACF) . . . 21
3.6 Diagnostic Tools: ACF and PACF . . . 21
3.6.1 ACF (Autocorrelation Function): The Total Correlation . . . 22
3.6.2 PACF (Partial Autocorrelation Function): The Direct Connection . . . 22
3.6.3 Summary Table for Model Identification . . . 22
3.7 Information Criteria: Choosing the Best Model . . . 22
3.8 The Connection: From Identification (ACF/PACF) to Selection (AIC/BIC) . . . 23
3.8.1 Step 1: The Detective Work (ACF and PACF) . . . 23
3.8.2 Step 2: The Competition (Estimation) . . . 23
3.8.3 Step 3: The Final Judge (AIC and BIC) . . . 24
3.8.4 Summary . . . 24

4 Part 4: ARIMA Models for Non-stationary Time Series **24**
4.1 The Concept of Non-stationarity . . . 24
4.1.1 The Random Walk (The Prototype of Non-stationarity) . . . 24
4.1.2 Theoretical Correlation and Smoothness (The Unity Property) . . . 25
4.1.3 Random Walk with Drift . . . 25
4.2 Integrated Processes: $I(d)$ . . . 25
4.2.1 The Difference Operator (Δ) . . . 25
4.2.2 Higher Order Differences ($I(2)$) . . . 25
4.3 The $ARIMA(p, d, q)$ Model . . . 25
4.3.1 General Equation . . . 26
4.4 Determining the Order of Integration (d) . . . 26
4.4.1 Difference between ACF and SACF . . . 26
4.4.2 Sample ACF (SACF) . . . 26
4.4.3 The Variance Rule . . . 27
4.4.4 Unit Root Tests (Formal Identification) . . . 27
4.4.5 Summary of Common d Values . . . 27
4.5 The Box-Jenkins Methodology . . . 27
4.6 Unit Root Testing: The Dickey-Fuller (DF) Logic . . . 27
4.6.1 1. The Basic Model and the Unit Root . . . 28
4.6.2 2. Why we rewrite the equation . . . 28
4.6.3 3. The Hypothesis Test . . . 28

4.6.4 4. The Dickey-Fuller Distribution 28
4.6.5 5. Augmented Dickey-Fuller (ADF) 28

5 Part 00: Neural Networks and Deep Learning 29
5.1 Foundations 29
5.2 Network Architecture 29
5.3 Training and Optimization 29

1 Part 1: Time Series and Their Features

1.1 Fundamental Definitions

A **time series** on a variable x is denoted as x_t, where t is the time index.

- **Observation Period:** $t = 1, 2, \ldots, T$, where T is the last observation.
- **Forecast Horizon:** Future values are denoted as x_{T+h}, where h is the horizon.
- **Sequential Order:** Unlike cross-sectional data, the order of observations in a time series is paramount because of potential temporal dependence.

1.2 Key Features of Time Series

1.2.1 Autocorrelation and Periodic Movements

Autocorrelation refers to the correlation of a series with its own past values. Many series exhibit periodic movements that are not strictly seasonal but repeat over certain intervals.

1.2.2 Seasonality

Seasonality consists of regular, short-term cycles that repeat within a fixed period, typically a year.

1.2.3 Trends

A trend represents the long-term evolution of a series.

- **Deterministic Trend:** Often modeled as a linear function of time: $x_t = \beta_0 + \beta_1 t + \epsilon_t$.
- **Stochastic Trend:** Represented by a "random walk" (explained afterwards) where the trend evolves based on cumulative random shocks.

1.2.4 Stationarity and Nonstationarity

A process is **stationary** if its statistical properties (mean, variance) remain constant over time. Stationarity formalizes the idea that the probabilistic behavior of a stochastic process does not change over time.

Strict (Strong) Stationarity

The process $\{X_t\}$ is **strictly stationary** if for every integer $k \geq 1$, for every collection of time points $t_1, \ldots, t_k$, and for every shift $h \in \mathbb{Z}$, the joint probability distributions respects:

$$P(X_{t_1}, \ldots, X_{t_k}) = P(X_{t_1+h}, \ldots, X_{t_k+h})$$

Weak (Second-Order) Stationarity

The process X_t is **weakly stationary** (or second-order stationary) if:

1. **The mean is constant:**
   $\mathbb{E}[X_t] = \mu$ for all t.
2. **The variance is finite and constant:**
   $\mathrm{Var}(X_t) = \gamma(0) < \infty$ for all t.
3. **The covariance depends only on the lag:**
   $\mathrm{Cov}(X_t, X_{t+h}) = \gamma(h)$ for all t, h.

Gaussian Process

A stochastic process X_t is a **Gaussian process** if for every finite collection of time points $t_1, \ldots, t_k$, the random vector

$$(X_{t_1}, \ldots, X_{t_k})$$

follows a multivariate Gaussian distribution.

For Gaussian processes:

$$\text{Weak stationarity} \iff \text{Strict stationarity}$$

This equivalence holds because Gaussian distributions are completely determined by their mean vector and covariance matrix.

Special case: White Noise

A stochastic process a_t is called **(weak) white noise** if:

- **Zero mean:** $\mathbb{E}[a_t] = 0$ for all t.
- **Constant finite variance:** $\mathrm{Var}(a_t) = \sigma^2$ for all t.
- **No serial correlation:**

$$\mathrm{Cov}(a_t, a_{t+h}) = \begin{cases} \sigma^2 & \text{if } h = 0, \\ 0 & \text{if } h \neq 0. \end{cases}$$

The process a_t is a weakly stationary process. In addition, if it is **Gaussian**, it holds that $\{a_t\}$ are independent and identically distributed (i.i.d.) with:

$$a_t \overset{i.i.d.}{\sim} N(0, \sigma^2)$$

Integration: A series is integrated of order d, denoted $I(d)$, if it must be differenced d times to become stationary.

First-Difference Operator (Δ): Used to transform a nonstationary series:

$$\Delta x_t = x_t - x_{t-1} \tag{1}$$

EWMA — Visual detection of non-stationarity in mean

A simple exploratory way to assess whether a time series is non-stationary in mean (i.e., presents a trend) is to plot a smoothed version of the series, obtained via an **exponentially weighted moving average (EWMA)**.

Given a series $\{X_t\}$, the exponentially smoothed mean is defined recursively as:

$$m_t = \alpha X_t + (1 - \alpha) m_{t-1}, \quad 0 < \alpha < 1,$$

or equivalently as the weighted average:

$$m_t = \sum_{k=0}^{\infty} \alpha (1 - \alpha)^k X_{t-k}.$$

Because recent observations receive more weight, m_t tracks the **local level** of the process. If $\{X_t\}$ is stationary in mean, m_t fluctuates around a constant; persistent drifts indicate the presence of a trend.

EWVAR — Visual detection of non-stationarity in variance

In the same spirit, we can visually inspect **stationarity in variance** by plotting a local variance estimate, specifically an exponentially weighted variance:

$$v_t = \alpha(X_t - m_t)^2 + (1 - \alpha)v_{t-1}.$$

If the process is stationary in variance, v_t should fluctuate around a constant level; systematic changes over time suggest **heteroskedasticity**.

Exemple

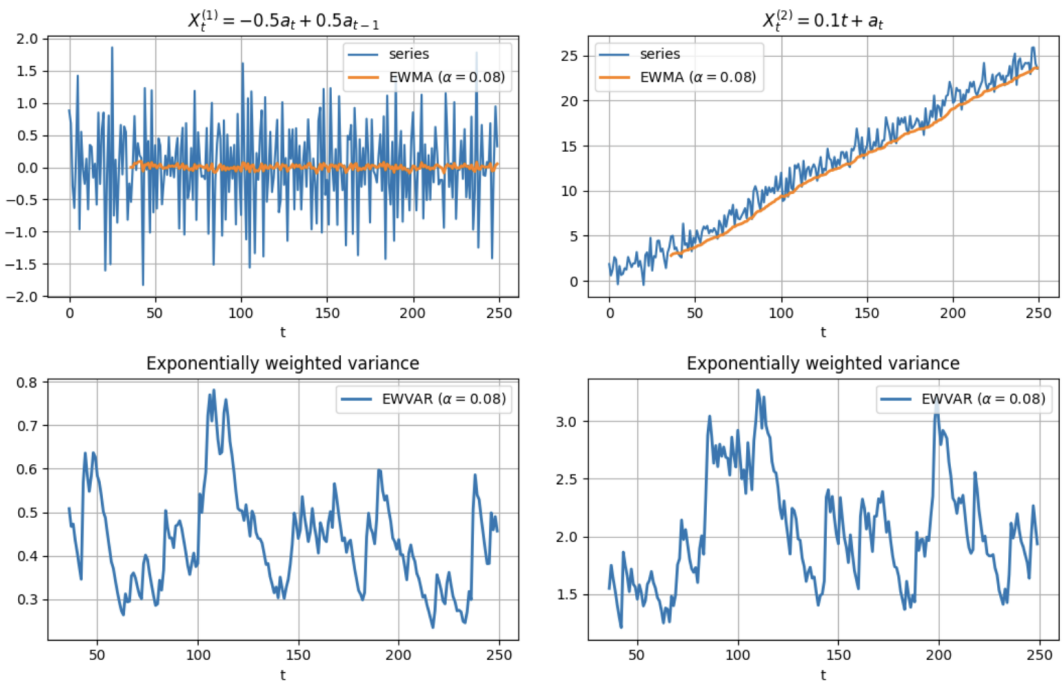


Figure 1: Exemple of EWMA and EWVAR plots

The provided plots illustrate the visual detection of non-stationarity using Exponentially Weighted Moving Average (EWMA) and Variance (EWVAR) with a smoothing parameter $\alpha = 0.08$.

Feature	**Process** $X_t^{(1)} = -0.5a_t + 0.5a_{t-1}$	**Process** $X_t^{(2)} = 0.1t + a_t$
Model Type	First-Order Moving Average, MA(1).	Trend-Stationary / Drift Process.
Mean Stationarity	**Stationary.** EWMA remains horizontal around zero; no time-dependent drift.	**Non-Stationary.** EWMA shows a clear diagonal slope due to the $0.1t$ term.
Variance Stationarity	**Stationary.** EWVAR fluctuations are bounded (0.3–0.8) with no systematic drift.	**Stationary.** EWVAR remains bounded (1.5–3.5); the "spread" is constant despite the trend.
Conclusion	Can be weakly stationary (if it is weakly stationary in autocovariance).	Non-stationary in mean, but stationary in variance. so Non weakly stationary in general.

Table 1: Comparative Analysis of Stationarity via EWMA and EWVAR

Key Takeaway

These simulations demonstrate that **stationarity in mean and variance are independent properties**. A series can exhibit a strong deterministic trend (non-stationary mean) while maintaining a stable internal noise structure (stationary variance).

Being stationary in mean and variance (and autocovariance) only satisfies Weak Stationarity. It does not guarantee Strict Stationarity.

In the world of statistics, EWMA and EWVAR are "diagnostic" or "exploratory" tools. They are excellent at falsifying stationarity, but they are insufficient for proving it.

1.2.5 Detrending via Iterated Differencing

When a time series exhibits a trend—meaning it is **non-stationary in mean**—a standard approach to remove the trend component is **iterated differencing**. This transformation produces a de-trended series that is often closer to being weakly stationary.

Given a process $\{X_t\}$, its **first difference** is defined as:

$$\nabla X_t = X_t - X_{t-1}$$

Heuristically, if the mean $\mathbb{E}[X_t]$ behaves like a polynomial trend of degree d (i.e., $\mathbb{E}[X_t] = O(t^d)$), then the d-th order difference, $\nabla^d X_t$, will have an **approximately constant mean**.

In practice:

- Applying the first difference removes a **linear trend** (like the $0.1t$ in your second graph).
- If the series still looks non-stationary (e.g., it had a quadratic trend), a **second difference** can be applied:

$$\nabla^2 X_t = \nabla(\nabla X_t) = (X_t - X_{t-1}) - (X_{t-1} - X_{t-2}) = X_t - 2X_{t-1} + X_{t-2}$$

1.2.6 Deterministic Trend Decomposition

Another common technique is to assume that the time series can be decomposed as:

$$X_t = f(t) + R_t$$

where $f(t)$ is a deterministic function representing the **trend** and R_t is a stationary process.

In this framework, non-stationarity is attributed entirely to the deterministic function $f(t)$, while the residual component R_t is assumed to be stationary (or at least closer to stationarity).

Polynomial Trend Approximation

A simple and widely used specification assumes that $f(t)$ is a polynomial of degree d:

$$f(t) = \beta_0 + \beta_1 t + \cdots + \beta_d t^d$$

The coefficients $\beta_0, \ldots, \beta_d$ are typically estimated via **Ordinary Least Squares (OLS)**. Once the trend $\hat{f}(t)$ is estimated, the de-trended series is obtained by analyzing the residuals:

$$\hat{R}_t = X_t - \hat{f}(t)$$

Example of Polynomial Detrending

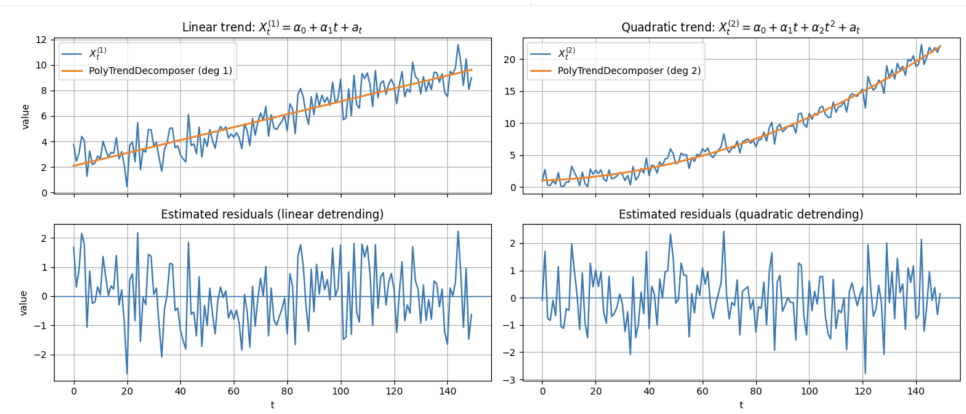


Figure 2: Polynomial Detrending

Feature	**Linear Case** ($X_t^{(1)}$)	**Quadratic Case** ($X_t^{(2)}$)
Mathematical Model	$X_t = \alpha_0 + \alpha_1 t + a_t$	$X_t = \alpha_0 + \alpha_1 t + \alpha_2 t^2 + a_t$
Visual Trend Shape	Constant upward slope (Linear).	Accelerating upward curve (Parabolic).
Decomposition Tool	PolyTrendDecomposer (**deg 1**)	PolyTrendDecomposer (**deg 2**)
Detrending Method	Subtraction of the fitted line.	Subtraction of the fitted curve.
Residual Analysis	Mean-stationary; noise centered at 0.	Mean-stationary; noise centered at 0.
Conclusion	Linear trend successfully removed.	Quadratic trend successfully removed.

Table 2: Comparison of Deterministic Polynomial Detrending Results

1.2.7 ACF and PACF

The **autocovariance function** is defined as:

$$\gamma(t, t+h) = \mathrm{Cov}(X_t, X_{t+h})$$

In general, $\gamma(t, t+h)$ may depend on both the time location t and the lag h.
The **autocorrelation function (ACF)** is:

$$\rho(t, t+h) = \frac{\gamma(t, t+h)}{\sqrt{\gamma(t,t)\gamma(t+h, t+h)}} = \mathrm{Corr}(X_t, X_{t+h})$$

Again, in general, it may depend on t.

Stationary Case

If X_t is **weakly stationary**, the autocovariance depends only on the lag h, and the notation simplifies to:

$$\gamma(h) = \mathrm{Cov}(X_t, X_{t+h}), \quad h \in \mathbb{Z}$$

Moreover, it satisfies the symmetry property: $\gamma(-h) = \gamma(h)$.
The autocorrelation function becomes:

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}$$

which is also symmetric: $\rho(-h) = \rho(h)$.

Partial Autocorrelation Function (PACF)

The **partial autocorrelation at lag** h is defined as:

$$\alpha(t, t+h) = \mathrm{Corr}(X_t, X_{t+h} \mid X_{t+1}, \ldots, X_{t+h-1})$$

This represents the correlation between X_t and X_{t+h} after removing the linear effect of the intermediate lags. For a weakly stationary process, this quantity depends only on the lag: $\alpha(h)$. Equivalently, $\alpha(h)$ is the last coefficient in the linear regression:

$$X_{t+h} = \phi_{h,1} X_{t+h-1} + \cdots + \phi_{h,h} X_t + \epsilon_{t+h}$$

so that $\alpha(h) = \phi_{h,h}$. As with the ACF, we have symmetry: $\alpha(-h) = \alpha(h)$.

Practical Considerations

The sample ACF and PACF are meaningful primarily for **stationary processes**.

- **Non-stationarity Signal:** If the series contains a deterministic trend, the sample ACF typically exhibits a **very slow decay** and remains close to 1 for many lags.
- **Diagnostic Utility:** A very slow decay in the sample ACF is a clear signal of non-stationarity, suggesting that transformations such as differencing or deterministic detrending are required before ARMA modeling.

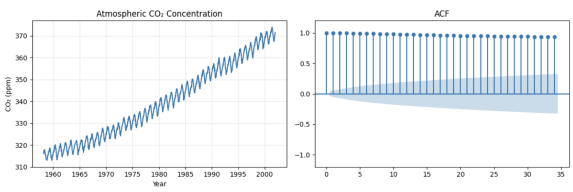


Figure 3: Raw Series

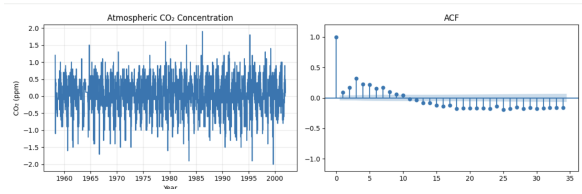


Figure 4: Transformed Series

Feature	Raw Series (Before ∇)	Transformed Series (After ∇)
Time Series Plot	Exhibits a strong, positive **deterministic trend** and a clear annual **seasonal pattern**. The increasing mean violates weak stationarity.	The global trend is removed. The series fluctuates around a mean of ≈ 0. The seasonal "sawtooth" pattern is now the dominant feature.
ACF Plot	**Extremely slow decay**; correlations remain near 1.0.	**Rapid decay** toward zero, a primary indicator of stationarity in mean.
Seasonal Signal	Masked by the overwhelming influence of the linear trend.	Periodic "waves" and significant spikes at regular intervals suggest **seasonal non-stationarity** remains.
Conclusion	Non-stationary in mean; requires transformation.	Mean-stationary regarding trend, but likely requires seasonal differencing (∇_{12}).

Table 3: Comparative Analysis of CO_2 Concentration: Raw vs. First-Differenced

1.2.8 Ergodicity in the Mean

A weakly stationary process is **ergodic in the mean** if time averages computed from a single long realization converge to the population mean. This property justifies using a single sample path to infer the characteristics of the entire stochastic process.

Definition

The sample mean is defined as:

$$\bar{X}_T = \frac{1}{T}\sum_{t=1}^{T} X_t$$

The process is ergodic in the mean if $\bar{X}_T \to \mu$ with probability 1 as $T \to \infty$. In practical terms, one sufficiently long realization contains all the information needed to estimate population quantities.

Statistical Properties

Unbiasedness:
For a weakly stationary process, the sample mean is an **unbiased** estimator of μ:

$$\mathbb{E}[\bar{X}_T] = \mu$$

Consistency:
Ergodicity requires **consistency**, meaning $\bar{X}_T \xrightarrow{a.s.} \mu$. Whether this holds depends on how quickly the temporal dependence (autocovariance) decays.

Condition on Autocovariance Decay

The variance of the sample mean is given by:

$$\mathrm{Var}(\bar{X}_T) = \frac{1}{T}\left(\gamma(0) + 2\sum_{h=1}^{T-1}\left(1 - \frac{h}{T}\right)\gamma(h)\right)$$

A sufficient condition for ergodicity (and thus consistency) is the **absolute summability** of the autocovariance function:

$$\sum_{h=-\infty}^{\infty} |\gamma(h)| < \infty$$

If this condition is satisfied, then $\mathrm{Var}(\bar{X}_T) \to 0$ as $T \to \infty$, ensuring that $\bar{X}_T \to \mu$ almost surely.

1.2.9 Seasonality and Frequency Domain Analysis

Seasonality can be detected, in addition to inspecting the time series plot, through a **periodic pattern in the ACF**. A seasonal process typically produces a sample ACF characterized by:

- Significant spikes at multiples of the seasonal period (e.g., $h = 12, 24, \ldots$ for monthly data).
- An **oscillatory or sinusoidal pattern in the autocorrelation function.**

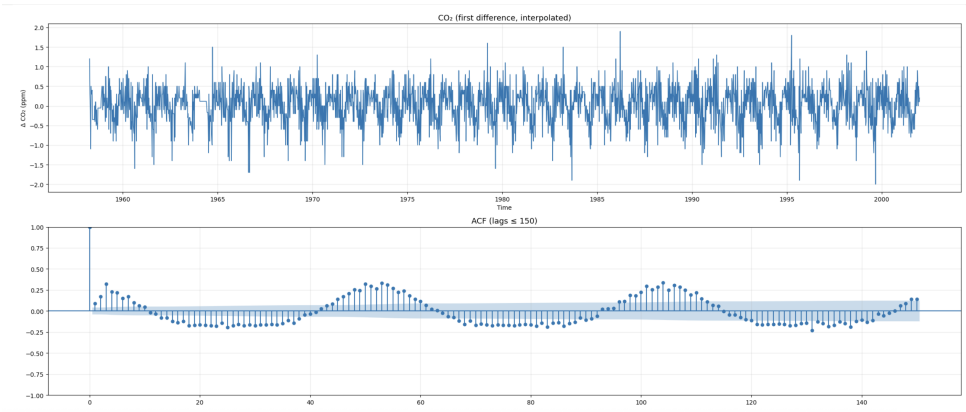


Figure 5: Example of CO2

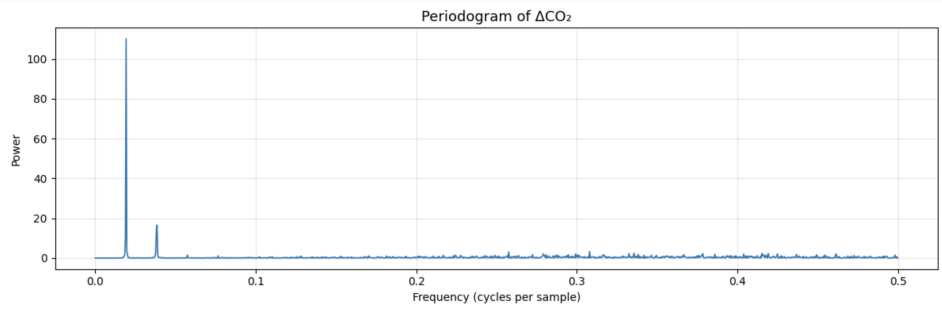
To further investigate periodic behavior, it is useful to analyze the **spectrum**, which provides a frequency-domain representation of the dependence structure.

The Periodogram

The periodogram provides an empirical estimate of how much power (variance) is present at each frequency ω_k. It is particularly useful for detecting **hidden periodic** components in the noise. It is computed as:

$$I(\omega_k) = \frac{1}{n}|X(\omega_k)|^2$$

where $X(\omega_k)$ is the **Discrete Fourier Transform (DFT)** of the data.




```
Top 10 spectral peaks:
 1) freq = 0.019273 cycles/sample | period = 51.89 samples | power = 110.108
 2) freq = 0.038546 cycles/sample | period = 25.94 samples | power = 16.5513
 3) freq = 0.038108 cycles/sample | period = 26.24 samples | power = 13.9266
 4) freq = 0.018835 cycles/sample | period = 53.09 samples | power = 10.0701
 5) freq = 0.019711 cycles/sample | period = 50.73 samples | power = 3.72012
 6) freq = 0.307928 cycles/sample | period = 3.25 samples | power = 3.25589
 7) freq = 0.257556 cycles/sample | period = 3.88 samples | power = 3.11238
 8) freq = 0.037670 cycles/sample | period = 26.55 samples | power = 2.45646
 9) freq = 0.335523 cycles/sample | period = 2.98 samples | power = 2.36213
10) freq = 0.415243 cycles/sample | period = 2.41 samples | power = 2.33497
```

Figure 6: Example of a Periodogram

1.2.10 Custom Spectrum: DCT-II with Standardization

To analyze the periodicity of the series, we employ a custom power spectrum approach based on the **DCT-II basis functions**.
For a signal of length n, the cosine basis used in the **DCT-II** is defined as:

$$\phi_i^{(k)} = \cos\left(\frac{\pi}{n}\left(i + \frac{1}{2}\right)k\right)$$

where:

- $i = 0, 1, \ldots, n-1$ is the **sample index**.
- $k = 0, 1, \ldots, n-1$ is the **basis index** (representing frequency).

Frequency and Period Relations

Each basis function k is characterized by its frequency and period in the time domain:

- **Angular Frequency:** $\omega_k = \frac{\pi k}{n}$ (radians/sample)
- **Sample Frequency:** $f_k = \frac{k}{2n}$ (cycles/sample)
- **Period:** $T_k = \frac{2\pi}{\omega_k} = \frac{2n}{k}$ (samples/cycle)

Note

As k increases, the frequency f_k increases and the period T_k decreases. Small values of k correspond to low-frequency components (like **trends**), while large values of k correspond to high-frequency components (like **noise**).

This method provides a frequency-domain representation of the energy distribution of the data energy distribution. The goal of this analysis is dimensionality reduction and modeling:

Identification of exactly which frequencies k capture the most variance in the data.

Instead of using the entire noisy signal, we can build a Fourier Series model using only the few k values with the highest power.

Standardization

Before computing the spectrum, the series x is standardized to ensure the visualization is scale-independent:

$$y = \frac{x - \mu}{\sigma}$$

where μ and σ are the mean and standard deviation of the series, respectively.

Note

Standardization is performed primarily for visualization purposes. Since it is a monotonic linear transformation, it **does not change the order relation** between the powers of different frequencies.

Power Spectrum Definition

We define the power spectrum sequence $P[k]$ as the squared norm of the **circular convolution** between the standardized series y and each cosine basis function $\phi^{(k)}$:

$$P[k] = \|\phi^{(k)} * y\|^2$$

where:

- $\phi^{(k)}$ is the DCT-II basis function of index k.
- $*$ denotes the circular convolution operator. between the standardized series y and the basis function $\phi^{(k)}$, both of length n, is defined as:

$$(y * \phi^{(k)})_i = \sum_{j=0}^{n-1} y_j \cdot \phi^{(k)}_{(i-j)} \pmod n$$

Property:

- This custom spectrum is that the functions $P[k]$ are **shift-invariant**.
- The circular convolution acts as a **matched filter**. It measures the correlation between the data and a cosine wave of frequency k, regardless of the wave's phase alignment

1.2.11 Deterministic Seasonality Decomposition

In this framework, we assume an additive decomposition model:

$$X_t = T(t) + S(t) + R_t$$

where:

- $T(t)$ is a deterministic, **aperiodic** function representing the trend (e.g., a polynomial).
- $S(t)$ is a deterministic **periodic** function representing seasonality.
- R_t is the stochastic residual, assumed to be approximately weakly stationary.

The objective is to explicitly model and remove the deterministic components $T(t)$ and $S(t)$ to isolate R_t for stationary time-series analysis.

Modeling Seasonality with a Truncated Fourier Series

A seasonal component with period s satisfies the condition $S_{t+s} = S_t$. A flexible and smooth method to represent such behavior is through a **truncated Fourier series**:

$$S_t = a_0 + \sum_{k=1}^{K} \left[a_k \cos\left(\frac{2\pi kt}{s}\right) + b_k \sin\left(\frac{2\pi kt}{s}\right) \right]$$

where:

- s is the seasonal period (e.g., $s = 12$ for monthly data).
- K is the number of harmonics retained (the "smoothness" of the fit).
- (a_k, b_k) are the coefficients to be estimated, typically via Ordinary Least Squares (OLS).

> **Tip:** Using a small K (e.g., $K = 1$ or 2) captures simple sinusoidal cycles, while increasing K allows the model to fit more complex, "sharper" seasonal patterns.

1.2.12 Distributional Analysis and Normality

Histograms for Visual Inspection of Normality

A common technique to assess normality is to compare the **histogram** of the empirical distribution of X_t with the **probability density function (PDF)** of a Gaussian distribution having the same mean and variance.

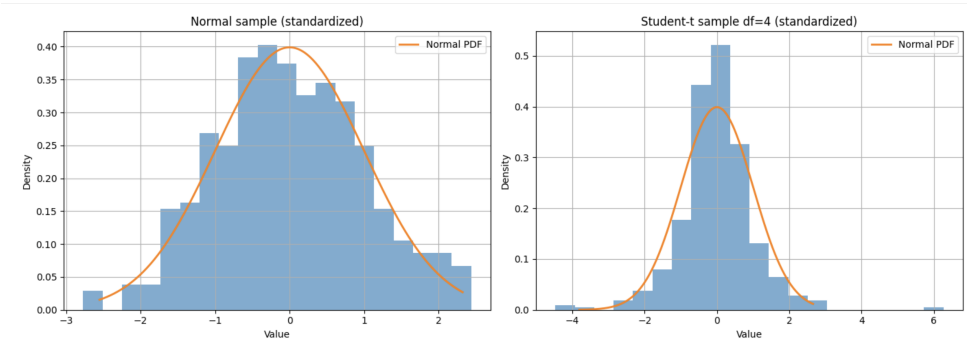


Figure 7: Gaussion Distibution

1.2.13 Last Example Of The lab 1 (see it)

Feature	Normal Distribution	Student-t ($df = 4$)
Visual Center	Matches the Normal PDF curve perfectly.	More peaked/concentrated at the mean than the Normal PDF.
Extreme Events	Extremely rare (tails decay very fast).	More frequent (leads to "Fat Tails").
Outliers	Values $> \|3\|$ are highly improbable.	Values like 6 (as seen in the plot) are possible.
Application	Used for standard "well-behaved" noise.	Used in finance/risk when "shocks" are more frequent.

Table 4: Comparison of Normal vs. Heavy-Tailed Distributions

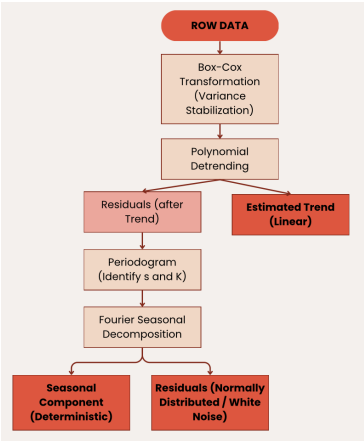


Figure 8: Example lab 1

1.2.14 Volatility

Volatility refers to a time-varying variance.

- **Volatility Clustering:** Periods of high variation followed by more high variation, common in financial data.
- **Model:** Often handled using GARCH (Generalized Autoregressive Conditional Heteroskedastic) models.

1.3 Natural Constraints

Some series must follow mathematical constraints due to their nature:

- **Count Data:** Observations are non-negative integers $(0, 1, 2 \dots)$.
- **Compositional Data:** Series representing shares of a whole, satisfying:

$$0 \leq x_{i,t} \leq 1 \quad \text{and} \quad \sum_{i=1}^{n} x_{i,t} = 1 \tag{2}$$

1.4 Multivariate Features

- **Common Trends:** Two or more nonstationary series may be "bound together" by a common underlying trend (Cointegration).
- **Causality:** One series may help predict another (Granger Causality).

2 Part 2: Transforming Time Series

Prior to statistical analysis, it is often necessary to transform data to make it more amenable to modeling. There are three general classes of transformations: distributional, stationarity-inducing, and decompositional.

2.1 Distributional Transformations

These aim to achieve approximate normality, symmetry, and constant variance.

2.1.1 Logarithmic Transformation

A popular choice for non-negative series that are skewed to the right. It "straightens out" exponential trends and stabilizes variance.

- **Usage:** Effective when the standard deviation is proportional to the level of the series.
- **Formula:** $y_t = \log(x_t)$.

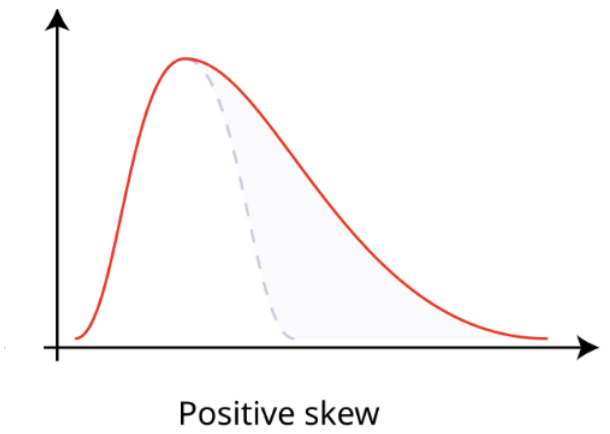


Figure 9: serie skewed to the right

2.1.2 Box-Cox Transformation

When a time series takes **strictly positive** values and shows **heteroskedasticity** (variance increasing with the level), a common choice is **the Box–Cox** transformation:

$$f_{BC}(x_t; \lambda) = \begin{cases} (x_t^\lambda - 1)/\lambda & \lambda \neq 0 \\ \log x_t & \lambda = 0 \end{cases} \tag{3}$$

In practice, specific values of λ correspond to common transformations:

- $\lambda = 1$: **No transformation** (the variance is already stable).
- $\lambda = 0$: **Logarithmic transformation** (used when variance increases linearly with the mean).
- $\lambda = 1/2$: **Square root transformation** (often used for count data).

Estimation

The parameter λ is typically chosen by **Maximum Likelihood Estimation (MLE)** to find the value that best transforms the data into a Gaussian-like distribution with constant variance.

2.1.3 Transformations for Proportions

When the observed process X_t represents a **proportion or probability** (where $0 \leq X_t \leq 1$), specific transformations are required to map the bounded interval to the real line $\mathbb{R}$ and stabilize the variance.

Logit Transformation

The most common approach is the **Logit transformation**:

$$Y_t = \ln\left(\frac{X_t}{1 - X_t}\right)$$

This transformation maps the domain $(0, 1)$ to $(-\infty, \infty)$. It is particularly effective for stabilizing variance in data that is "squeezed" near 0 or 1.

Boundary Corrections

Since the logit function is undefined for $X_t = 0$ or $X_t = 1$, a small correction is often applied to the data before transformation. A common adjustment is:

$$X_t^* = \frac{X_t(n-1) + 0.5}{n}$$

where n is the sample size. This ensures all values are strictly within the interior of the unit interval $(0, 1)$, allowing the logit calculation to proceed.

> **Tip:** The logit transformation is effectively the "inverse" of the sigmoid function used in logistic regression. It turns an S-shaped curve into a linear one.

2.1.4 Handling Negative Values

For series with negative values, several alternatives exist:

- **Signed Power:** $f_{SP}(x_t; \lambda) = \text{sgn}(x_t)|x_t|^\lambda - 1/\lambda$ for $\lambda > 0$.
- **Inverse Hyperbolic Sine (IHS):** Useful for extreme values of either sign:

$$f_{IHS}(x_t; \lambda) = \frac{\sinh^{-1}(\lambda x_t)}{\lambda} = \frac{\log(\lambda x_t + (\lambda^2 x_t^2 + 1)^{1/2})}{\lambda} \tag{4}$$

2.2 Stationarity Inducing Transformations

To remove deterministic or stochastic trends, differencing is applied.

2.2.1 First Differencing

Used for series that are integrated of order 1 ($I(1)$):

$$\Delta x_t = (1 - B)x_t = x_t - x_{t-1} \tag{5}$$

where B is the backshift operator ($Bx_t = x_{t-1}$).

2.2.2 Seasonal Differencing

For a series with a seasonal cycle of length s (e.g., $s = 12$ for monthly data):

$$\Delta_s x_t = (1 - B^s)x_t = x_t - x_{t-s} \tag{6}$$

2.2.3 The Shift Operator

The **Shift Operator** S^k (also known as the Lag Operator L^k) is a linear operator that shifts the elements of a time series. For a finite vector x of length n, the *circular shift* is defined as:

$$(S^k x)_i = x_{i-k \pmod n}$$

Key Properties

- **Linearity:** $S^k(\alpha x + \beta y) = \alpha S^k x + \beta S^k y$.
- **Composition:** $S^j S^k = S^{j+k}$.
- **Inverse:** S^{-k} is the forward shift operator
- The Periodogram is **shift-invariant**.

> **Tip:** In time series decomposition, the shift operator is used to define filters. For example, a moving average is simply a weighted sum of shifted versions of the signal.

2.3 Decomposing a Time Series

Observed series X_t are often decomposed into Trend (T_t), Seasonal (S_t), and Irregular (I_t) components.

2.3.1 Additive vs. Multiplicative Models

- **Additive:** $X_t = T_t + S_t + I_t$.
- **Multiplicative:** $X_t = T_t \times S_t \times I_t$ (often transformed to additive via logs).

2.3.2 Smoothing Transformations (Moving Averages)

A simple way to extract the trend is via a symmetric $2k+1$ point moving average:

$$M_t = \frac{1}{2k+1} \sum_{j=-k}^{k} x_{t+j} \tag{7}$$

2.3.3 Hodrick-Prescott (HP) Filter

A more advanced decomposition that minimizes the following sum for a chosen smoothing parameter λ:

$$\min \sum_{t=1}^{T} (x_t - T_t)^2 + \lambda \sum_{t=2}^{T-1} [(T_{t+1} - T_t) - (T_t - T_{t-1})]^2 \tag{8}$$

Common values for λ include 14,400 for monthly data or 1,600 for quarterly data.

3 Part 3: ARMA Models for Stationary Time Series

3.1 Intuitive Interpretation of AR and MA Components

Before diving into the complex estimations, it is helpful to understand the behavioral "personality" of these models.

3.1.1 AR (Autoregressive): The Long-Term Memory

The **AR** component describes a system that depends on its **own previous values**.

- **Analogy:** It is like a person's mood; if someone was happy yesterday, he is likely to still be somewhat happy today.
- **Mechanism:** Each new value is a weighted percentage of the last value, plus a current "random surprise."
- **Impact:** It creates **momentum** or **persistence**. For example, if the temperature is 30°C today, it is unlikely to drop to 0°C tomorrow because the "memory" of today's heat lingers.

3.1.2 MA (Moving Average): The Shock Absorber

The **MA** component does **not** depend on the previous values of the series, but rather on the **previous random shocks** (surprises or errors).

- **Analogy:** Imagine an earthquake. The "shock" happens today. Tomorrow, we feel an "aftershock." By the third day, that specific shock is gone.
- **Mechanism:** It looks at the "random noise" from yesterday and determines how much of that noise should still ripple into today.
- **Impact:** It handles **short-term echoes**. Once the shock exceeds the order q, it is completely forgotten.

3.1.3 ARMA: The Combined Personality

Combining them into an **ARMA** model allows us to represent a series with:

- **Persistence (AR):** The tendency to stay where it was.
- **Reaction to Shocks (MA):** The tendency to ripple for a short time after a sudden surprise.

3.1.4 Summary Comparison

Feature	AR (Autoregressive)	MA (Moving Average)
Logic	"I am where I am because of where I was."	"I am where I am because of the 'news' I just received."
Memory	Long-term (fades geometrically).	Short-term (cuts off abruptly).
Analogy	**Momentum:** Like a car rolling after we stop hitting the gas.	**Echo:** Like a sound bouncing for a moment, then vanishing.

3.2 Theoretical Foundations

3.2.1 The Stochastic Process

A **time series** is a realization of a **stochastic process** $\{x_t, t \in \mathbb{Z}\}$. We assume the process is **linearly regular**, meaning it can be expressed as a linear combination of its past innovations.

3.2.2 Stationarity (Covariance Stationarity)

A process is weakly stationary if:

1. $E[x_t] = \mu$ (The mean is constant for all t).
2. $Var(x_t) = E[(x_t - \mu)^2] = \sigma^2 = \gamma_0 < \infty$.
3. $Cov(x_t, x_{t-k}) = E[(x_t - \mu)(x_{t-k} - \mu)] = \gamma_k$ for all t.

The **Autocorrelation Function (ACF)** is defined as:

$$\rho_k = \frac{\gamma_k}{\gamma_0} \tag{9}$$

Note that $\rho_0 = 1$ and $|\rho_k| \leq 1$.

3.2.3 The White Noise Process

The building block of all ARMA models is **White Noise** $\{a_t\}$, defined as:

- $E[a_t] = 0$
- $E[a_t^2] = \sigma_a^2$
- $E[a_t a_s] = 0$ for $t \neq s$ (uncorrelated shocks).

3.3 Autoregressive Processes: $AR(p)$

3.3.1 The $AR(1)$ Model: Detailed Derivation

The model is $x_t = \phi x_{t-1} + a_t$.
By repeated substitution (assuming the process started in the infinite past):

$$x_t = a_t + \phi a_{t-1} + \phi^2 a_{t-2} + \cdots = \sum_{j=0}^{\infty} \phi^j a_{t-j} \tag{10}$$

Variance calculation:

$$\gamma_0 = Var\left(\sum_{j=0}^{\infty} \phi^j a_{t-j}\right) = \sigma_a^2 \sum_{j=0}^{\infty} \phi^{2j} = \frac{\sigma_a^2}{1 - \phi^2} \tag{11}$$

This requires the **Stationarity Condition**: $|\phi| < 1$.

3.3.2 The General $AR(p)$ and Yule-Walker Equations

The $AR(p)$ model is $x_t = \phi_1 x_{t-1} + \cdots + \phi_p x_{t-p} + a_t$.
To find the autocorrelations, we multiply by x_{t-k} and take expectations. This yields the **Yule-Walker Equations**:

$$\rho_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2} + \cdots + \phi_p \rho_{k-p} \quad \text{for } k > 0 \tag{12}$$

This system of equations allows us to solve for the parameters ϕ_i if we know the autocorrelations ρ_k.

3.4 Moving Average Processes: $MA(q)$

3.4.1 The $MA(1)$ Model

Defined as $x_t = a_t - \theta a_{t-1}$.

- **Mean:** $E[x_t] = E[a_t] - \theta E[a_{t-1}] = 0$.
- **Variance:** $\gamma_0 = E[(a_t - \theta a_{t-1})^2] = \sigma_a^2(1+\theta^2)$.
- **Autocovariance at lag 1:** $\gamma_1 = E[(a_t - \theta a_{t-1})(a_{t-1} - \theta a_{t-2})] = -\theta\sigma_a^2$.
- **ACF:** $\rho_1 = \frac{-\theta}{1+\theta^2}$ and $\rho_k = 0$ for $k > 1$.

3.4.2 Invertibility

An $MA(1)$ is **invertible** if $|\theta| < 1$.
If invertible, we can express the error a_t as an infinite sum of past observations:

$$a_t = \sum_{j=0}^{\infty} \theta^j x_{t-j} \tag{13}$$

3.4.3 General $MA(q)$ Autocorrelation

For a general $MA(q)$ model $x_t = a_t - \theta_1 a_{t-1} - \cdots - \theta_q a_{t-q}$, the autocorrelation at lag k is calculated as:

$$\rho_k = \begin{cases} \frac{-\theta_k + \theta_1\theta_{k+1} + \cdots + \theta_{q-k}\theta_q}{1+\theta_1^2+\cdots+\theta_q^2} & k = 1, \ldots, q \\ 0 & k > q \end{cases} \tag{14}$$

3.5 The $ARMA(p,q)$ Process

3.5.1 General Definition

Combining both parts:

$$(1 - \phi_1 B - \cdots - \phi_p B^p)x_t = (1 - \theta_1 B - \cdots - \theta_q B^q)a_t \tag{15}$$

Where B is the **Backshift Operator** ($B^k x_t = x_{t-k}$).
This is often written as $\Phi(B)x_t = \Theta(B)a_t$.

3.5.2 The Partial Autocorrelation Function (PACF)

While the ACF of an AR model tails off (Gradual Decay), the **PACF** "cuts off" (Abrupt Stop)).
The PACF at lag k (denoted ϕ_{kk}) is the correlation between x_t and x_{t-k} after controlling for the effects of $x_{t-1}, \ldots, x_{t-k+1}$.

- For an $AR(p)$, $\phi_{kk} = 0$ for $k > p$.
- For an $MA(q)$, the PACF tails off (opposite of the ACF).

3.6 Diagnostic Tools: ACF and PACF

To decide whether a series is an AR, an MA, or a mix of both ($ARMA$), we use two primary functions that measure how observations are related to each other over time.

3.6.1 ACF (Autocorrelation Function): The Total Correlation

The **ACF** at lag k, denoted as ρ_k, measures the simple correlation between x_t and its past value x_{t-k}. It is defined as the ratio of the autocovariance to the variance:

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \frac{E[(x_t - \mu)(x_{t-k} - \mu)]}{E[(x_t - \mu)^2]} \tag{16}$$

- **Behavior for MA:** It is the "ID card" for Moving Average models. For an $MA(q)$, the ACF will be significant for q lags and then suddenly drop to zero (**cuts off**).
- **Behavior for AR:** For Autoregressive models, the ACF **tails off** (fades slowly) because the memory of the past persists through a chain reaction.
  For an $AR(1)$, this decay is geometric: $\rho_k = \phi^k$.

3.6.2 PACF (Partial Autocorrelation Function): The Direct Connection

The **PACF** at lag k, denoted as ϕ_{kk}, measures the correlation between x_t and x_{t-k} **after removing the effects** of all intervening observations $(x_{t-1}, \ldots, x_{t-k+1})$. It is formally the last coefficient of an $AR(k)$ regression:

$$x_t = \phi_{k1}x_{t-1} + \phi_{k2}x_{t-2} + \cdots + \phi_{kk}x_{t-k} + a_t \tag{17}$$

To compute ϕ_{kk} from the autocorrelations ρ_k, we use:

$$\phi_{11} = \rho_1, \quad \phi_{kk} = \frac{\rho_k - \sum_{j=1}^{k-1} \phi_{k-1,j}\rho_{k-j}}{1 - \sum_{j=1}^{k-1} \phi_{k-1,j}\rho_j} \tag{18}$$

- **Logic:** It reveals if x_{t-k} has a *direct* impact on x_t.
- **Behavior for AR:** It is the "ID card" for AR models.
  In an $AR(1)$, the PACF at lag 1 is ϕ, but for all lags $k > 1$, the PACF is 0.
  For an $AR(p)$, the PACF will be significant for p lags and then suddenly **cut off**.
- **Behavior for MA:** For Moving Average models, the PACF **tails off** (fades slowly).

3.6.3 Summary Table for Model Identification

This table is used by analysts to "read" the plots and choose the model order (p, q):

Model	**ACF Plot**	**PACF Plot**
AR(p)	Tails off (geometric decay)	**Cuts off after lag** p
MA(q)	**Cuts off after lag** q	Tails off (geometric decay)
ARMA(p,q)	Tails off	Tails off

3.7 Information Criteria: Choosing the Best Model

When we build a model, we face a trade-off: a complex model fits the current data well but may fail to forecast the future (**over-fitting**), while a simple model might miss important patterns. We use Information Criteria to find the "sweet spot" by minimizing the following:

- **AIC (Akaike Information Criterion):** $AIC = -2\ln(L) + 2k$. It provides a small penalty for each parameter (k). It is often used when the primary goal is high-quality forecasting.

- **BIC (Bayesian Information Criterion):** $BIC = -2\ln(L) + k\ln(T)$. It provides a much larger penalty that increases with the sample size (T). It is "conservative," meaning it prefers simpler models with fewer parameters to ensure the model is truly representative of the underlying process.

L **(Likelihood):** This refers to the **Maximum Likelihood**. It represents how well the model explains the observed data.

- In the criteria equations, the term $-2\ln(L)$ measures the "misfit" or unexplained information.
- As the model fit improves, L increases, causing $-2\ln(L)$ to decrease (reducing the overall "cost").

k **(Number of Parameters):** This represents the **complexity** of the model.

- For an $ARIMA(p, d, q)$ model, $k = p + q + 1$ (where $+1$ typically accounts for the constant/intercept or the noise variance).
- This serves as the **penalty term**. Adding AR or MA terms to "force" a better fit increases k, thereby increasing the penalty.

T **(Sample Size):** This is the total number of **observations** in the time series.

- In the BIC formula, the penalty is $k\ln(T)$.
- As the dataset grows (T increases), the BIC becomes significantly more "strict" than the AIC regarding the addition of new parameters.

Rule of Thumb: Calculate both for various combinations of p and q. The model with the **lowest score** is generally considered the best balance of accuracy and simplicity.

3.8 The Connection: From Identification (ACF/PACF) to Selection (AIC/BIC)

In practice, the process of building an ARMA model follows a specific logical sequence. The Information Criteria (AIC/BIC) act as the final decision-makers when the visual tools are ambiguous.

3.8.1 Step 1: The Detective Work (ACF and PACF)

We use the **ACF** and **PACF** to "shortlist" potential models:

- If the PACF cuts off at lag 2, we suspect an **AR(2)**.
- If the ACF cuts off at lag 1, we suspect an **MA(1)**.
- **The Problem:** In real-world data, the plots are rarely perfect. we might see patterns that look like they could be an $AR(2)$, an $MA(2)$, or an $ARMA(1, 1)$ all at once. This is where we need a tie-breaker.

3.8.2 Step 2: The Competition (Estimation)

We estimate all the "suspect" models identified in Step 1. We now have three different versions of reality:

1. Model A: $AR(2)$ (Uses 2 parameters: ϕ_1, ϕ_2)
2. Model B: $MA(2)$ (Uses 2 parameters: θ_1, θ_2)
3. Model C: $ARMA(1, 1)$ (Uses 2 parameters: ϕ_1, θ_1)

3.8.3 Step 3: The Final Judge (AIC and BIC)

Since all these models might have a similar number of parameters (k), we use **AIC** and **BIC** to see which one explains the data most efficiently.

- **Why use them with AR/MA?** If add more lags (e.g., jumping from an $AR(2)$ to an $AR(10)$), our model will always fit the training data better, but it becomes "noisy."
- **The Penalty:** AIC and BIC "punish" the model for every extra p (AR part) or q (MA part) we add.
- **Selection:** we calculate the AIC/BIC for all candidates. The one with the **lowest value** is the winner. It tells us exactly how much "memory" (AR) and how many "shocks" (MA) are truly necessary without overcomplicating the model.

3.8.4 Summary

- **PACF/ACF:** Tell us which *direction* to look in (Is it memory-based or shock-based?).
- **AIC/BIC:** Tell us exactly where to *stop* adding complexity (How many lags are enough?).

4 Part 4: ARIMA Models for Non-stationary Time Series

4.1 The Concept of Non-stationarity

Most real-world time series (like GDP, stock prices, or consumer indices) do not have a constant mean. They exhibit trends or "drifts."

4.1.1 The Random Walk (The Prototype of Non-stationarity)

A process is a **Random Walk** if the current value is the previous value plus a random shock (White Noise WN):

$$x_t = x_{t-1} + a_t, \quad a_t \sim WN(0, \sigma_a^2) \tag{19}$$

By back-substitution, assuming $x_0 = 0$:

$$x_t = a_t + a_{t-1} + a_{t-2} + \cdots + a_1 = \sum_{i=1}^{t} a_i \tag{20}$$

Properties of a Random Walk:

- **Mean:** $E[x_t] = 0$ (if no drift).
- **Variance (γ notation):** Since $Var(a_i) = \sigma_a^2$, the variance at time t is:

$$\gamma_{0;t} = Var(x_t) = t\sigma_a^2 \tag{21}$$

- **Autocovariance:** For $k > 0$, the covariance between x_t and its past x_{t-k} is:

$$\gamma_{k;t} = Cov(x_t, x_{t-k}) = (t-k)\sigma_a^2 \tag{22}$$

- **Conclusion:** Because $\gamma_{0;t}$ depends on time t (it grows to infinity), the process is **non-stationary**.

4.1.2 Theoretical Correlation and Smoothness (The Unity Property)

The autocorrelation between x_t and x_{t-k} for a random walk is given by:

$$\rho_{k,t} = \frac{\gamma_{k,t}}{\sqrt{\gamma_{0,t}\gamma_{0,t-k}}} = \frac{(t-k)\sigma_a^2}{\sqrt{(t\sigma_a^2)((t-k)\sigma_a^2)}} = \sqrt{\frac{t-k}{t}} \tag{23}$$

Crucial Observation: If t is large compared to k, then $\rho_{k,t} \approx 1$.
Implication: The sequence of x_t values will appear very **smooth** because consecutive values are almost perfectly correlated, yet the series remains **non-stationary** as the variance continues to explode ($t\sigma_a^2 \to \infty$).

4.1.3 Random Walk with Drift

If we add a constant μ, we get a **Random Walk with Drift**:

$$x_t = \mu + x_{t-1} + a_t \implies x_t = x_0 + \mu t + \sum_{i=1}^{t} a_i \tag{24}$$

This produces a **stochastic trend** where the "level" of the series changes by μ on average each period.

4.2 Integrated Processes: $I(d)$

A series x_t is said to be **integrated of order** d, denoted $x_t \sim I(d)$, if it must be differenced d times to become stationary.

4.2.1 The Difference Operator (Δ)

The first difference is defined using the Backshift Operator B:

$$\Delta x_t = (1-B)x_t = x_t - x_{t-1} \tag{25}$$

If x_t is a Random Walk ($I(1)$), then its first difference $\Delta x_t = a_t$ is White Noise (Stationary).

4.2.2 Higher Order Differences ($I(2)$)

For some series, the "slope" itself is changing. We then use the second difference:

$$\Delta^2 x_t = \Delta(\Delta x_t) = (1-B)^2 x_t = (1-2B+B^2)x_t = x_t - 2x_{t-1} + x_{t-2} \tag{26}$$

The "slope" refers to the rate of change between consecutive observations.
For example: In equation (26) The "slope" is μ.
If a series has a constant slope, it follows a straight line (a deterministic trend).
If the slope is "changing," it means the series is curving or accelerating, which usually indicates the series is Integrated of order 2, denoted as $I(2)$.

4.3 The $ARIMA(p,d,q)$ Model

The **Autoregressive Integrated Moving Average** model combines **differencing** with the ARMA structure.

4.3.1 General Equation

A process x_t is an $ARIMA(p, d, q)$, if $w_t = \Delta^d x_t$ is a **stationary** $ARMA(p, q)$ process:

$$\Phi(B)(1 - B)^d x_t = \Theta(B)a_t \tag{27}$$

Expanding the operators:

$$(1 - \phi_1 B - \cdots - \phi_p B^p)(1 - B)^d x_t = (1 - \theta_1 B - \cdots - \theta_q B^q)a_t \tag{28}$$

4.4 Determining the Order of Integration (d)

The goal of the identification stage is to find the minimum integer d such that $w_t = \Delta^d x_t$ is stationary. We follow these specific technical steps:

4.4.1 Difference between ACF and SACF

In time series analysis, it is crucial to distinguish between the theoretical properties of a process and the empirical evidence we calculate from data.

- **ACF (Autocorrelation Function):** This is a **theoretical property** of a model. It describes the population-level relationship between observations.
  - *Example:* If we **know** a process is exactly an $AR(1)$ with $\phi = 0.8$, we can use mathematical derivation to determine that $\rho_1 = 0.8$, $\rho_2 = 0.64$, and so on.
  - It assumes we either have an infinite amount of data or are the "creator" of the process and know the true parameters.
- **SACF (Sample Autocorrelation Function):** This is an **empirical estimate** denoted as $\hat{\rho}_k$.
  - In reality, the true parameters (like ϕ) are unknown. we only possess a finite sample of T observations.
  - The SACF is the statistical tool used to estimate what the underlying theoretical ACF looks like based on the available data.

4.4.2 Sample ACF (SACF)

The **SACF** ($\hat{\rho}_k$) is the empirical estimate of the autocorrelation at lag k for a series of size T. It is calculated as:

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^{T}(x_t - \bar{x})(x_{t-k} - \bar{x})}{\sum_{t=1}^{T}(x_t - \bar{x})^2} \tag{29}$$

It is the primary tool for determining the order of an $MA(q)$ process (where it cuts off) and detecting non-stationarity (where it decays very slowly).

- **Non-stationary indication:** If the SACF ($\hat{\rho}_k$) decays *very slowly* or remains high for many lags, the series is non-stationary ($d \geq 1$).
- **Stationary indication:** If the SACF drops to zero exponentially or cuts off quickly, the current level of differencing is sufficient.

4.4.3 The Variance Rule

A common rule of thumb is to compare the variance of the series at different levels of differencing. We choose the d that **minimizes the variance**:

$$Var(\Delta^d x_t) < Var(\Delta^{d-1} x_t) \tag{30}$$

If differencing again *increases* the variance, we have likely "over-differenced" the series.

4.4.4 Unit Root Tests (Formal Identification)

To move beyond visual patterns, we use formal statistical tests like the **Augmented Dickey-Fuller (ADF)** test:

- **Null Hypothesis (H_0):** The series has a unit root ($\phi = 1$), meaning it is non-stationary ($d \geq 1$).
- **Alternative Hypothesis (H_1):** The series is stationary ($d = 0$).

If we fail to reject H_0, we increment d and test the differenced series Δx_t until stationarity is confirmed (p-value < 0.05).

4.4.5 Summary of Common d Values

- $d = 0$: Series is stationary around a constant mean (e.g., White Noise, ARMA).
- $d = 1$: Series has a stochastic trend (e.g., Stock prices, Random Walk).
- $d = 2$: Series has an accelerating trend or a changing slope (rare in economics, except for hyper-inflation data).

4.5 The Box-Jenkins Methodology

The course emphasizes a three-step iterative cycle for fitting ARIMA models:

1. **Identification:**
   - Determine the order of integration d (usually by looking if the ACF decays very slowly, indicating non-stationarity).
   - Use ACF/PACF of the *differenced* series to pick p and q.
2. **Estimation:** Use Maximum Likelihood Estimation (MLE) or Least Squares to find the numerical values of ϕ_i and θ_j.
3. **Diagnostic Checking:** Ensure the residuals $\hat{a}_t$ behave like **White Noise**. If there is still a pattern in the residuals, the model is incomplete. Check using the **Ljung-Box test**:

$$Q = T(T+2) \sum_{k=1}^{m} \frac{\hat{\rho}_k^2(\hat{a})}{T-k} \sim \chi^2(m-p-q) \tag{31}$$

4.6 Unit Root Testing: The Dickey-Fuller (DF) Logic

The Dickey-Fuller test is the formal statistical procedure used to determine the order of integration d. It answers the question: "Is the series a Random Walk ($I(1)$) or is it mean-reverting ($I(0)$)?"

4.6.1 1. The Basic Model and the Unit Root

Consider a simple $AR(1)$ process:

$$x_t = \phi x_{t-1} + a_t \tag{32}$$

The parameter ϕ determines the persistence of the series:

- If $|\phi| < 1$: The series is **Stationary**. Shocks eventually vanish.
- If $\phi = 1$: The series has a **Unit Root**. It is a **Random Walk**. Shocks stay in the system forever (infinite memory).

4.6.2 2. Why we rewrite the equation

In standard regression, we test if a coefficient is 0. However, testing if $\phi = 1$ directly using standard t-tests is mathematically problematic because the variance of a Random Walk is not constant.

To fix this, we subtract x_{t-1} from both sides to create a **Difference Equation**:

$$x_t - x_{t-1} = \phi x_{t-1} - x_{t-1} + a_t \tag{33}$$

$$\Delta x_t = (\phi - 1)x_{t-1} + a_t \tag{34}$$

We then define a new parameter $\gamma = (\phi - 1)$. The model becomes:

$$\Delta x_t = \gamma x_{t-1} + a_t \tag{35}$$

4.6.3 3. The Hypothesis Test

By switching to γ, we can now test for 0 instead of 1:

- **Null Hypothesis (H_0):** $\gamma = 0$ (This implies $\phi = 1$).
  $\rightarrow$ The process has a **Unit Root** (Non-stationary).
- **Alternative Hypothesis (H_1):** $\gamma < 0$ (This implies $\phi < 1$).
  $\rightarrow$ The process is **Stationary**.

4.6.4 4. The Dickey-Fuller Distribution

Crucially, under H_0, the ratio $\frac{\hat{\gamma}}{SE(\hat{\gamma})}$ (where SE is the Standard Error) does **not** follow a standard Student's t-distribution. It follows the **Dickey-Fuller Distribution**.

- If the calculated t-statistic is **more negative** than the critical value from the DF table, we reject H_0 and conclude the series is stationary ($d = 0$).
- If we fail to reject, we assume $d \geq 1$, difference the data, and potentially test again.

4.6.5 5. Augmented Dickey-Fuller (ADF)

In reality, the residuals a_t might be autocorrelated (not white noise). To fix this, we "augment" the equation by adding lags of the dependent variable (Δx_{t-j}):

$$\Delta x_t = \alpha + \beta t + \gamma x_{t-1} + \delta_1 \Delta x_{t-1} + \cdots + \delta_p \Delta x_{t-p} + a_t \tag{36}$$

This ensures that the test accounts for higher-order $AR(p)$ dynamics while still focusing on the coefficient γ to check for the unit root.

5 Part 00: Neural Networks and Deep Learning

Based on the `neuralnet.pdf` lecture, this represents a non-linear approach to regression and time series forecasting.

5.1 Foundations

- **Definition:** A neural network is a non-linear parametric regression model formed by the repeated composition of linear functions (weights W and bias b) and non-linear **Activation Functions** $\sigma(\cdot)$.
- **Activation Functions:** Common choices include **Sigmoid**, **ReLU** (Rectified Linear Unit), and **Tanh**.
- **Kolmogorov's Mapping Theorem:** States that any continuous function can be approximated by a composition of simpler functions, justifying the depth of neural architectures.

5.2 Network Architecture

- **Feed-Forward Networks:** Information flows strictly in one direction: Input $\mathbf{x} \to$ Hidden Layers $\mathbf{h} \to$ Output $\mathbf{y}$.
- **Hidden Layer Mapping:** $\mathbf{h} = \sigma(\mathbf{W}\mathbf{x} + \mathbf{b})$.
- **Convolutional Neural Networks (CNN):** Optimized for grid-like data using the **Convolution Operation**:

$$[w * a]_i = \sum_k w_k a_{i-k+1} \tag{37}$$

  where w is the **kernel** (filter) that extracts spatial features.

5.3 Training and Optimization

- **Backpropagation:** The recursive application of the chain rule to compute the gradient of the loss function with respect to every weight in the network.
- **Error Decomposition:**
  - **Approximation Error:** Loss due to the restricted complexity of the chosen model class.
  - **Estimation Error:** Loss due to finite sample size and the presence of noise (leads to overfitting if too high).